

Pizza Sales SQL Analysis

Basic Analysis

```
SELECT COUNT(DISTINCT order_id) AS total_orders FROM orders;
```

```
SELECT SUM(od.quantity * p.price) AS total_revenue
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id;
```

```
SELECT pizza_id, price
FROM pizzas
ORDER BY price DESC
LIMIT 1;
```

```
SELECT p.size, COUNT(*) AS count
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY count DESC
LIMIT 1;
```

```
SELECT pt.name, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

Intermediate Analysis

```
SELECT pt.category, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category;
```

```
SELECT HOUR(time) AS order_hour, COUNT(*) AS total_orders
FROM orders
GROUP BY order_hour
ORDER BY order_hour;
```

```
SELECT pt.category, COUNT(*) AS total_orders
FROM pizza_types pt
JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od ON p.pizza_id = od.pizza_id
GROUP BY pt.category;
```

```
SELECT AVG(daily_count) AS avg_pizzas_per_day
FROM (
    SELECT o.date, SUM(od.quantity) AS daily_count
    FROM orders o
    JOIN order_details od ON o.order_id = od.order_id
    GROUP BY o.date
) AS daily_orders;
```

```
SELECT pt.name, SUM(od.quantity * p.price) AS revenue
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

Advanced Analysis

```
SELECT pt.name,  
SUM(od.quantity * p.price) AS revenue,  
(SUM(od.quantity * p.price) /  
  (SELECT SUM(od.quantity * p.price)  
    FROM order_details od  
    JOIN pizzas p ON od.pizza_id = p.pizza_id)) * 100 AS percentage  
FROM order_details od  
JOIN pizzas p ON od.pizza_id = p.pizza_id  
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id  
GROUP BY pt.name  
ORDER BY percentage DESC;
```

```
SELECT o.date,  
SUM(SUM(od.quantity * p.price)) OVER (ORDER BY o.date) AS cumulative_revenue  
FROM orders o  
JOIN order_details od ON o.order_id = od.order_id  
JOIN pizzas p ON od.pizza_id = p.pizza_id  
GROUP BY o.date;
```

```
SELECT category, name, revenue  
FROM (  
  SELECT pt.category, pt.name,  
  SUM(od.quantity * p.price) AS revenue,  
  RANK() OVER (PARTITION BY pt.category ORDER BY SUM(od.quantity * p.price) DESC) AS rank  
  FROM order_details od  
  JOIN pizzas p ON od.pizza_id = p.pizza_id  
  JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id  
  GROUP BY pt.category, pt.name  
) ranked  
WHERE rank <= 3;
```